

ELEC2302/9302 Signals and Systems — Quiz 1 formula sheet

Complex exponentials

$$e^{j\theta} = \cos \theta + j \sin \theta$$

A complex number in Cartesian and polar form:

$$z = x + jy = r e^{j\theta}, \quad r = |z| = \sqrt{x^2 + y^2}, \quad \theta = \arg z.$$

Inner product, energy and power

$$\langle u, y \rangle := \int_{-\infty}^{\infty} \bar{u}(t) y(t) dt$$

$$E(u) = \int_{-\infty}^{\infty} |u(t)|^2 dt, \quad P(u) = \lim_{W \rightarrow \infty} \frac{1}{2W} \int_{-W}^W |u(t)|^2 dt, \quad u_{\text{RMS}} = \sqrt{P(u)}.$$

If u is periodic with period T , then

$$\langle u, y \rangle_P = \frac{1}{T} \int_0^T \bar{u}(t) y(t) dt, \quad P(u) = \langle u, u \rangle_P = \frac{1}{T} \int_0^T |u(t)|^2 dt.$$

Standard signals

Unit step (Heaviside).

Pulse of width T .

$$H(t) = \int_{-\infty}^t \delta(\varsigma) d\varsigma = \begin{cases} 0, & t < 0 \\ 1, & t \geq 0 \end{cases}$$

$$p_T(t) = \begin{cases} 1, & |t| \leq T/2 \\ 0, & \text{otherwise} \end{cases}$$

$$p_T(t) = H\left(t + \frac{T}{2}\right) - H\left(t - \frac{T}{2}\right), \quad \frac{d}{dt}H(t) = \delta(t).$$

The Dirac delta

Sifting property.

Scaling.

$$\int_{-\infty}^{\infty} \delta(t - \tau) y(t) dt = y(\tau).$$

$$\delta(\alpha t) = \frac{1}{|\alpha|} \delta(t).$$

Convolution

$$(h * u)(t) := \int_{-\infty}^{\infty} h(t - \tau) u(\tau) d\tau = (u * h)(t)$$

$$(u * \delta)(t) = u(t), \quad (u * \delta(t - T))(t) = u(t - T), \quad h_1 * (h_2 * h_3) = (h_1 * h_2) * h_3.$$

Impedances

$$Z_R = R, \quad Z_L = j\omega L, \quad Z_C = \frac{1}{j\omega C}.$$